



Laxmi Charitable Trust's
Sheth L.U.J. College of Arts & Sir M.V. College of Science & Commerce
Dr. S. Radhakrisnan Marg, Andheri (East) Mumbai 400 069



Department of Information Technology (Bsc.IT Semester III)
Python Programming Practical

Practical – 6

Practical No. 6 - Write Algorithm, Code and Output

Roll No. : 34	Name: Mehek Sawant
Class : SY-IT	Date/Time of Submission: 28-07-2026

a. Write a program to compute the number of characters and words in a string.

Code:

```
s="Second year IT students"
```

```
word_count=len(s.split())
```

```
char_count=len(s)
```

```
#printing result
```

```
print("The number of the words in a string are :"+str(word_count))
```

```
print("The number of characters a string are :",char_count)
```

Output:

```
The number of the words in a string are :4
The number of characters a string are : 23
```

b. Create a file geometry.py to calculate base areas for shapes square and circle. In another file, write a function pointyShapeVolume(x, y, squareBase) that calculates

the volume of a square pyramid if squareBase is True and of a right circular cone if squareBase is False. x is the length of an edge on a square if squareBase is True and the radius of a circle when squareBase is False. y is the height of the object. First use squareBase to distinguish the cases. Use the circleArea and squareArea from the geometry module to calculate the base areas.

Code:

```
- main.py
import geometry

def pointyShapeVolume(x, y, squareBase):
    if squareBase:
        baseArea = geometry.squareArea(x)
    else:
        baseArea = geometry.circleArea(x)

    volume = (1/3) * baseArea * y
    return volume

print("Square Pyramid Volume =", pointyShapeVolume(4, 6, True))
print("Cone Volume =", pointyShapeVolume(3, 5, False))

- geometry.py
import math

def squareArea(x):
    return x**2

def circleArea(r):
    return math.pi*r**2
```

Output:

```
= RESTART: C:/Users/DELL/OneDrive/Desk  
Python/main.py  
Square Pyramid Volume = 32.0  
Cone Volume = 47.12388980384689  
|
```